Embedded System Lab Manual

Subject Name: Embedded System Lab

Subject Code: 7IT06

Experiment No.: 6

Title: Temperature Measurement and Display Using 4-Pin NTC Thermistor Temperature Sensor Module and Arduino UNO with RG1602A LCD Display

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Objectives

- To interface a 4-pin NTC thermistor temperature sensor module with Arduino UNO.
- To measure temperature using the thermistor sensor.
- To display the temperature reading on the RG1602A LCD display.
- To understand the working principle of NTC thermistor and LCD interfacing.

Required Components:

- Arduino UNO
- 4-Pin NTC Thermistor Temperature Sensor Module
- RG1602A LCD Display
- Breadboard
- Connecting Wires
- USB Cable for Arduino UNO

Required Libraries

- **LiquidCrystal** (Usually pre-installed with Arduino IDE)
  To include the library, use: #include <LiquidCrystal.h>

Theory

NTC Thermistor:

An NTC (Negative Temperature Coefficient) thermistor decreases its resistance as temperature increases. The thermistor is part of a voltage divider circuit which allows Arduino to read an analog voltage that corresponds to temperature.

LCD RG1602A:

The RG1602A is a 16x2 character LCD display module commonly used for displaying alphanumeric characters. It communicates with the Arduino through digital pins using 4-bit or 8-bit data modes.

Working Principle:

- The thermistor module outputs an analog voltage corresponding to temperature.
- Arduino reads the analog input voltage.
- Using the thermistor resistance-temperature characteristics and calibration, Arduino converts the analog value to temperature.
- The temperature is then displayed on the LCD.

Procedure

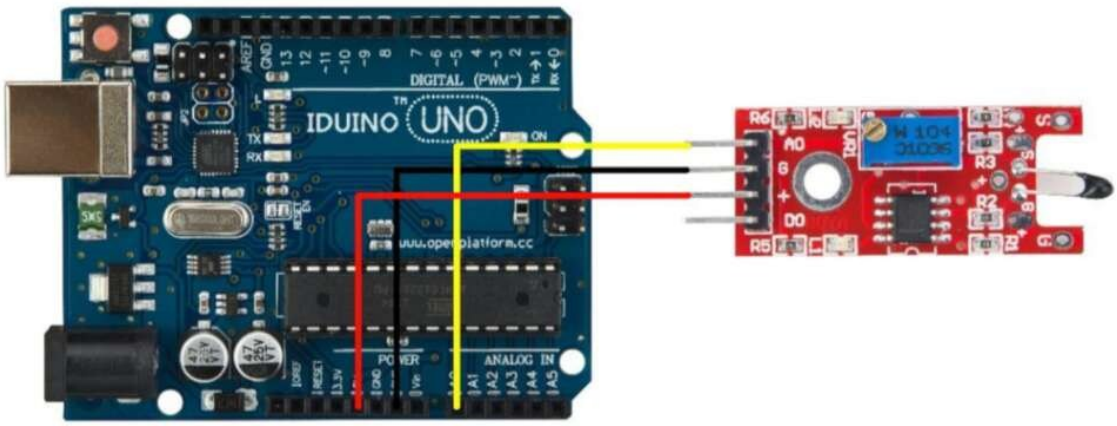
1. **Setup the hardware:**
   - Connect the 4-pin NTC thermistor module to Arduino (VCC, GND, Analog output pin to Arduino analog input).
   - Connect the RG1602A LCD to Arduino (RS, EN, D4, D5, D6, D7 pins) with appropriate power and ground connections.
2. **Upload the code:**
   - Open Arduino IDE, write or paste the provided code.
   - Upload the code to Arduino UNO.
3. **Observe:**
   - The LCD should display the temperature reading in °C.
4. **Test:**
   - Vary temperature by touching or exposing the sensor to different environments.
   - Note the changes on the LCD.

Circuit Diagram Description

Component Connection	Arduino Pin
Thermistor VCC	5V

Component Connection	Arduino Pin
Thermistor GND	GND
Thermistor Analog Output	A0
LCD VSS	GND
LCD VDD	5V
LCD VO (Contrast)	Potentiometer wiper or fixed resistor (e.g., 10kΩ to GND)
LCD RS	Pin 12
LCD EN	Pin 11
LCD D4	Pin 5
LCD D5	Pin 4
LCD D6	Pin 3
LCD D7	Pin 2
LCD RW	GND

Note: Use a potentiometer to adjust the contrast of the LCD.



Code

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);      // Initialize the LCD with the numbers of the interface pins
const int thermistorPin = A0;
const int seriesResistor = 10000;           // 10k Ohm resistor in series with thermistor
const int nominalResistance = 10000;        // resistance of thermistor at 25°C
const int nominalTemperature = 25;          // nominal temperature value
const int betaCoefficient = 3950;           // Beta coefficient of the thermistor
const int samples = 5;                      // Number of samples for averaging

void setup()
{
        lcd.begin(16, 2);
        lcd.print("Temp: ");
        delay(2000);
}
void loop()
{
        float average = 0;
                                            // Take multiple samples for averaging
        for (int i = 0; i < samples; i++)
        {
                average += analogRead(thermistorPin);
                delay(10);
        }
        average /= samples;
                                            // Convert the value to resistance
        float resistance = seriesResistor / ((1023.0 / average) - 1);
                                            // Calculate temperature using Steinhart-Hart equation
        float steinhart;
        steinhart = resistance / nominalResistance;          // (R/Ro)
        steinhart = log(steinhart);                          // ln(R/Ro)
        steinhart /= betaCoefficient;                        // 1/B * ln(R/Ro)
```

```
    steinhart += 1.0 / (nominalTemperature + 273.15);   // + (1/To)
    steinhart = 1.0 / steinhart;                        // Invert
    steinhart -= 273.15;                                // convert to °C

    lcd.clear();                                        // Display temperature on LCD
    lcd.setCursor(0, 0);
    lcd.print("Temp:");
    lcd.setCursor(6, 0);
    lcd.print(steinhart, 1);                            // 1 decimal place
    lcd.print(" C");

    delay(1000);
}
```

Observation Table

Sample Number	Calculated Temperature (°C)
1	
2	
3	
4	

Result

The Arduino UNO successfully reads the analog voltage from the 4-pin NTC thermistor sensor module, converts it into temperature in Celsius using the Steinhart-Hart equation, and displays it on the RG1602A LCD.

Conclusion

- The experiment demonstrated how to interface a 4-pin NTC thermistor temperature sensor module with Arduino.
- The temperature readings were accurately displayed on the LCD in real-time.
- This setup can be used for basic temperature monitoring systems.
- Understanding the thermistor characteristics and calibration is crucial for accurate temperature measurement.